DELFT UNIVERSITY OF TECHNOLOGY
Faculty of Electrical Engineering, Mathematics and Computer Science

CSE2315 Lab Course
Solutions 7

deadline: March 26, 2026, 13:45

1. Prove that the following problem is in NP:

$$\{(V, E) \mid \text{The graph } G = (V, E) \text{ contains a simple path } \pi \text{ such that } \forall v \in V : v \in \pi\}$$

In a different notation we would write:

Name: HAMILTONIAN CYCLE (HAMC)

Input: A graph $G = (V, E)$.

Question: Is there a simple cycle π in G such that all vertices $v \in V$ are in that cycle?

Solution: Consider the following algorithm, given a graph $G = (V, E)$ and a cycle π.

(a) Check if every vertex $v \in V$ occurs in π exactly once, except for the first which is also the last.

(b) Check for every consecutive pair of vertices in $(v_i, v_{i+1}) \in \pi$ that $(v_i, v_{i+1}) \in E$.

The first step takes $|V| \times |\pi|$ time, the second takes $|\pi| \times |E|$ time. Since the input $I = (V, E, \pi)$, this makes the time complexity $O(|I|^2)$ thus the problem can be verified in polynomial time, thus the problem is in NP.

2. *Old exam question, endterm 23-24:*

For the following claim, if it is true: prove it. If they are false, give an explanation showing the claim is false and if possible a counterexample. Remember that the cover of our exam reads: "Unless otherwise stated we assume that P $\neq$ NP and that NP $\neq$ EXP."

For three languages A, B, C if $A \leq_P B$ and $B \leq_m C$ with $C \in$ NPC. Furthermore we know that there is a regular expression R such that $L(R) = A$. Then $B \in$ P.

Solution: False, consider $A = \Sigma^*, B = C = \texttt{VC}$. Clearly $B \leq_m C$ just have a reduction that does nothing. Furthermore $A \in$ P so also in NP and thus it is Karp-reducible to VC which is NPC. Yet VC is not in P unless P = NP.

3. (Old exam question) On the (famous?) British Quiz Show "The University Challenge", the following question was asked in the grand final of 2018/2019.

> [...] which problem in computing asks whether all problems with a solution verifiable in finite time, can also be solved in finite time.

The contestants answer with "P equals NP" and the host answers back with "Yes that's right, P versus NP, yes".

Assuming this is the answer the host was looking for, what is wrong with the question he asks?

Solution: P and NP is not about finite time, but polynomial time! In finite time both can be done.

4. *Old exam question, resit 20-21:*

(a) Prove that the following problem is in NP.

Name AMAZINGCUSTOMCHALLENGE

Instance Given a set L of lectures, a set of students S, a collection of subsets T with $|T| = |S|$ so that T_i is the set of lectures that student i needs to attend, and an integer K.

Question Is there a function $f : L \to \{1, 2, \ldots, K\}$ that maps lectures to time slots so that all students can attend all their lectures. In other words is there a function so that there is no student who has to attend two lectures that are mapped to the same time slot.

> **Solution:** Given a function f, check first that every element from L is mapped $(O(|L|))$ correctly to a slot between 1 and K; second, check that for every subset that all exams are mapped to unique time slots $(O(|S||L|))$. This is polnomial in the input size.

(b) *Not originally part of the exam question* Your answer to the previous question should have included time complexities. If we are to describe them in terms of $|I|$ where I is an instance of the problem, what would be the complexity? (Hint: first think about $|I|$, is it $|S|$, or $|L|$, or some combination? How does K factor into this?)

> **Solution:** $|I| = |S| + |T| \cdot |L| + \log K$. So this is linear in $|I|$ since $|I|$ already contains an $|S| \cdot |L|$ term.

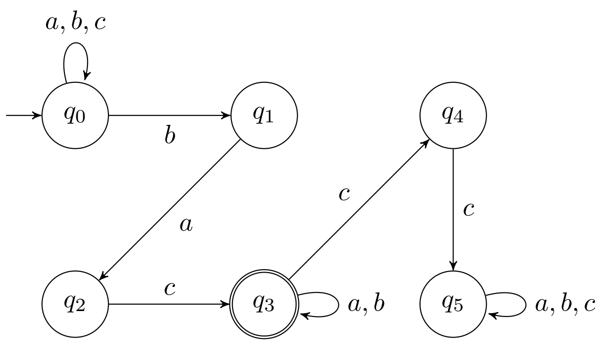
5. *Variation on old exam question, resit 20-21:*

   Show the claim is true or false, starting your answer with either "True" or "False".

   Given two problems A and B that can be solved using a PDA, it is possible to create a Karp-reduction from A to B and vice versa.

   > **Solution:**
   >
   > False, the empty set and sigma star are a counterexample. We cannot map a problem with both yes- and no-instances to either of these, but they are both context-free.

6. *Revision, old exam quesiton, resit 20-21:* Given this NFA N, give a regular expression R so that $L(N) = L(R)$.



   > **Solution:** $R = (a \cup b \cup c)^* bac(a \cup b)^*$

7. *Revision, old exam question, endterm 16-17:* Suppose we have the following language:

$$L = \{\langle M, v, w\rangle \mid M \text{ outputs } w \text{ on the tape when run on input } v\}$$

   Use a mapping reduction to prove that L is undecidable. Give:

   (a) A suitable problem to reduce from.

Solution: The problem used here is $HALT_{\mathsf{TM}}$.

(b) A suitable reduction function $f : \Sigma^* \to \Sigma^*$.

Solution: The reduction f is computed by the following TM F:
$F =$"On input $\langle M, w\rangle$:

1 Construct the following TM M':
$M' =$"On input x:

1 Clear the input from the tape.
2 Run M on w.
3 Write an a to the tape and accept."

2 Write $\langle M', a, a\rangle$ to the output tape."

(c) A proof showing f satisfies the requirements of a mapping reduction.

Solution:

Proof. We need to show that f is computable and reliable. F computes f, so f is computable. f is also reliable, because:

- $\langle M, w\rangle \in HALT_{\mathsf{TM}} \Rightarrow f(\langle M, w\rangle) \in L$ holds, since if M halts on w, M' leaves a on the tape on any input x and this is exactly what it is supposed to leave. So, $\langle M', a, a\rangle \in L$.
- $\langle M, w\rangle \notin HALT_{\mathsf{TM}} \Rightarrow f(\langle M, w\rangle) \notin L$ holds, since if M does not halt on w, M' will never write the correct output to the tape.

$\square$

End of the assignment